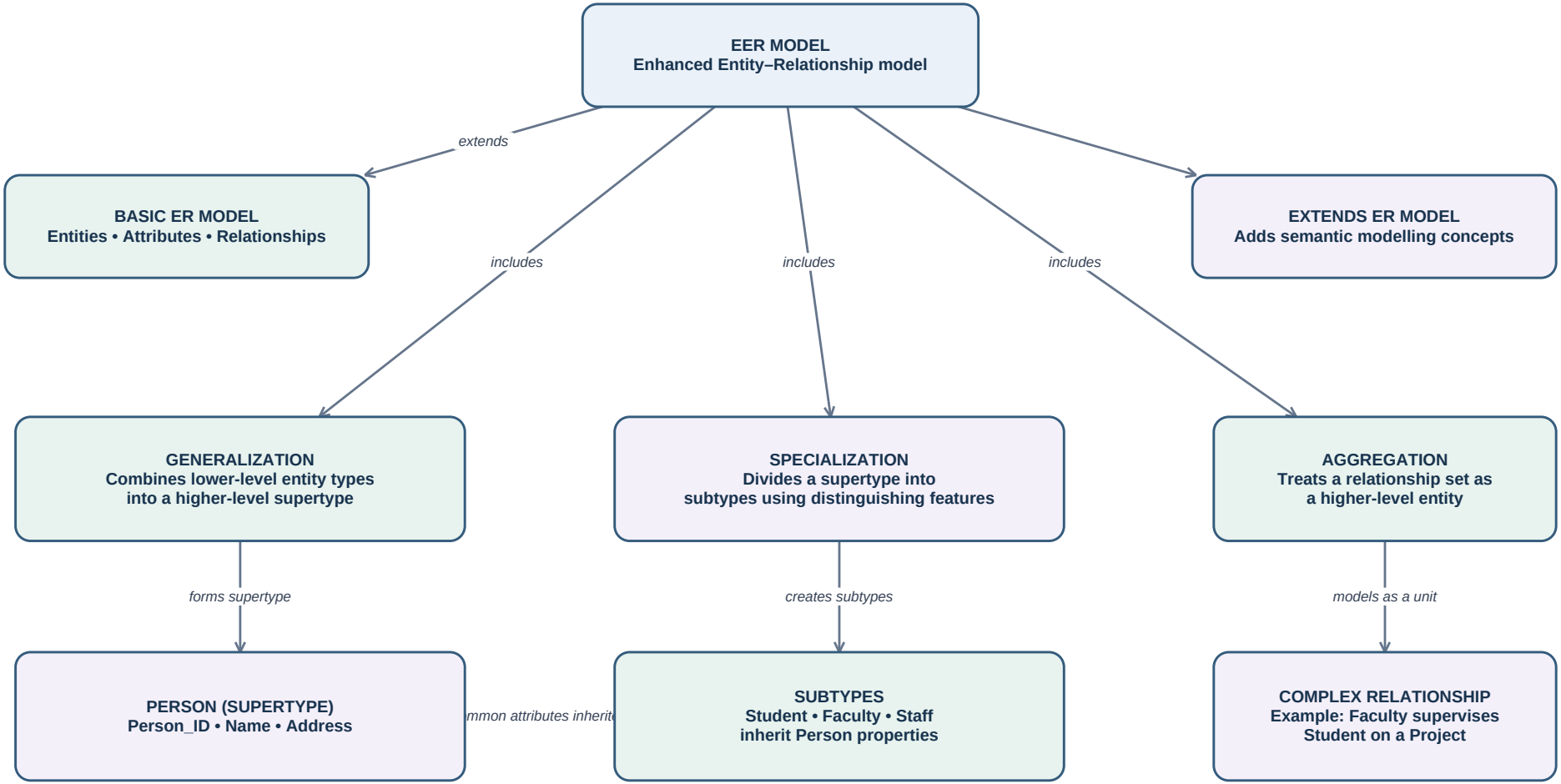
# Q1. EER MODEL: GENERALIZATION, SPECIALIZATION & AGGREGATION

How the Enhanced ER model extends the basic ER model in a university database

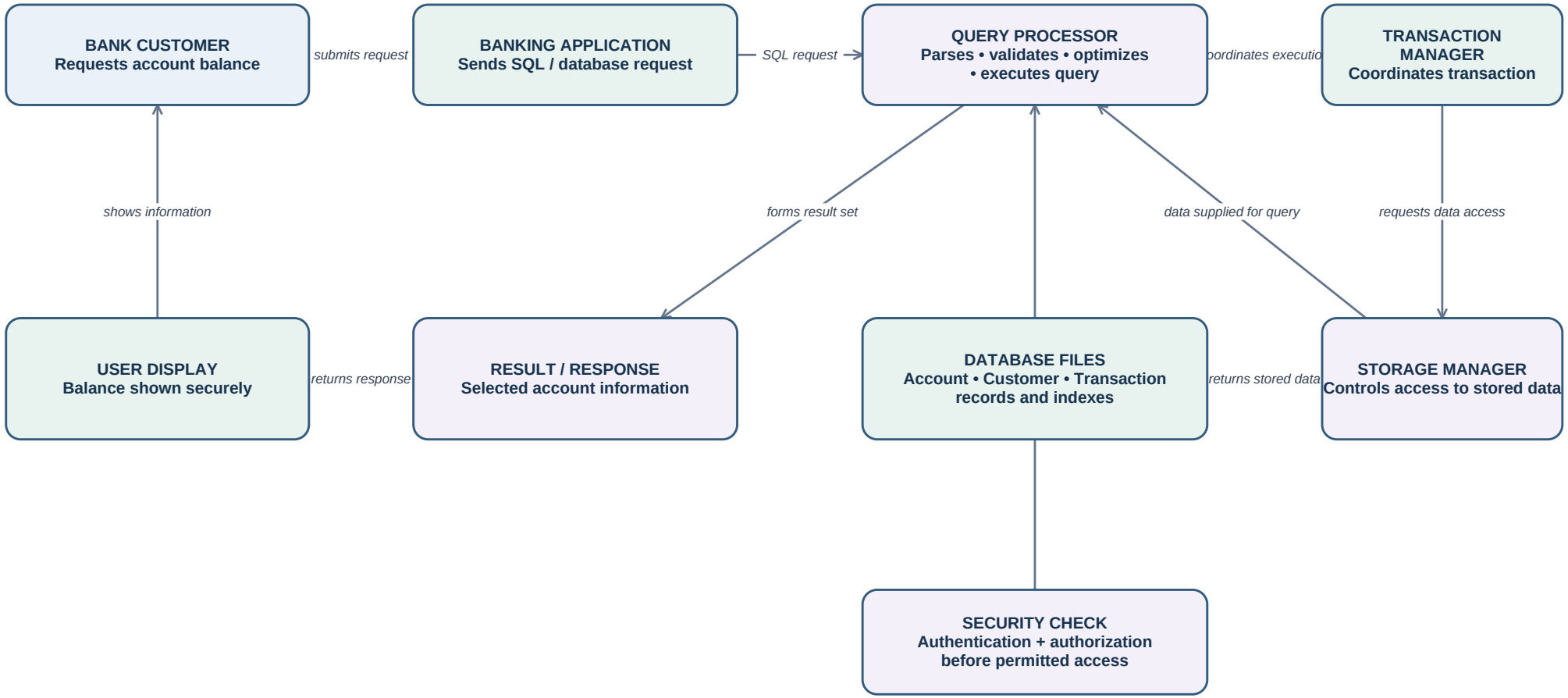



**How to explain the map:** Key relationships: Generalization is bottom-up (Student, Faculty and Staff → Person). Specialization is top-down (Person → Student, Faculty and Staff). In this example, the subtypes inherit Person_ID, Name and Address and add their own attributes. Aggregation abstracts a relationship so it can participate in another relationship; for example, a Faculty member supervises a Student's work on a Project. Generalization/specialization may be subject to disjoint/overlapping and total/partial constraints where specified.

**Presentation tip:** Follow the arrow direction, read each linking phrase, and explain why the example fits the concept. Use the same labels when writing or presenting the map.

# Q2. DBMS SOFTWARE ARCHITECTURE: ONLINE BANKING REQUEST

Flow of a user request to retrieve account information, from application to stored data and back

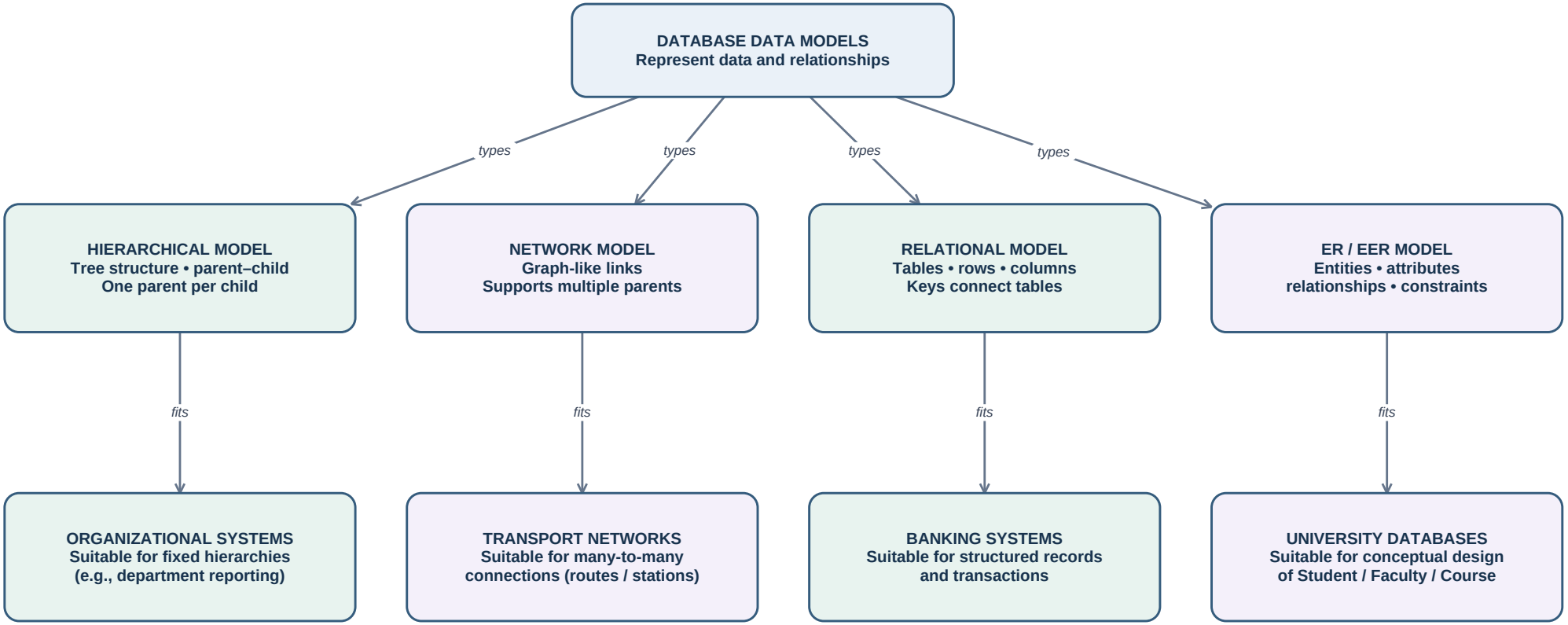



**How to explain the map:** Read the main flow from left to right across the top, then down through storage and back toward the user. The query processor interprets and executes the query; the transaction manager coordinates safe transaction processing; and the storage manager handles files, indexes, buffers and access to stored data. The result is returned to the application for display. Authentication and authorization are security checks that must be applied before account data is disclosed. The exact internal order can vary by DBMS implementation.

**Presentation tip:** Follow the arrow direction, read each linking phrase, and explain why the example fits the concept. Use the same labels when writing or presenting the map.

# Q3. DATA MODELS AND THEIR REAL-WORLD SUITABILITY

Compare the structure each model represents and where it is useful

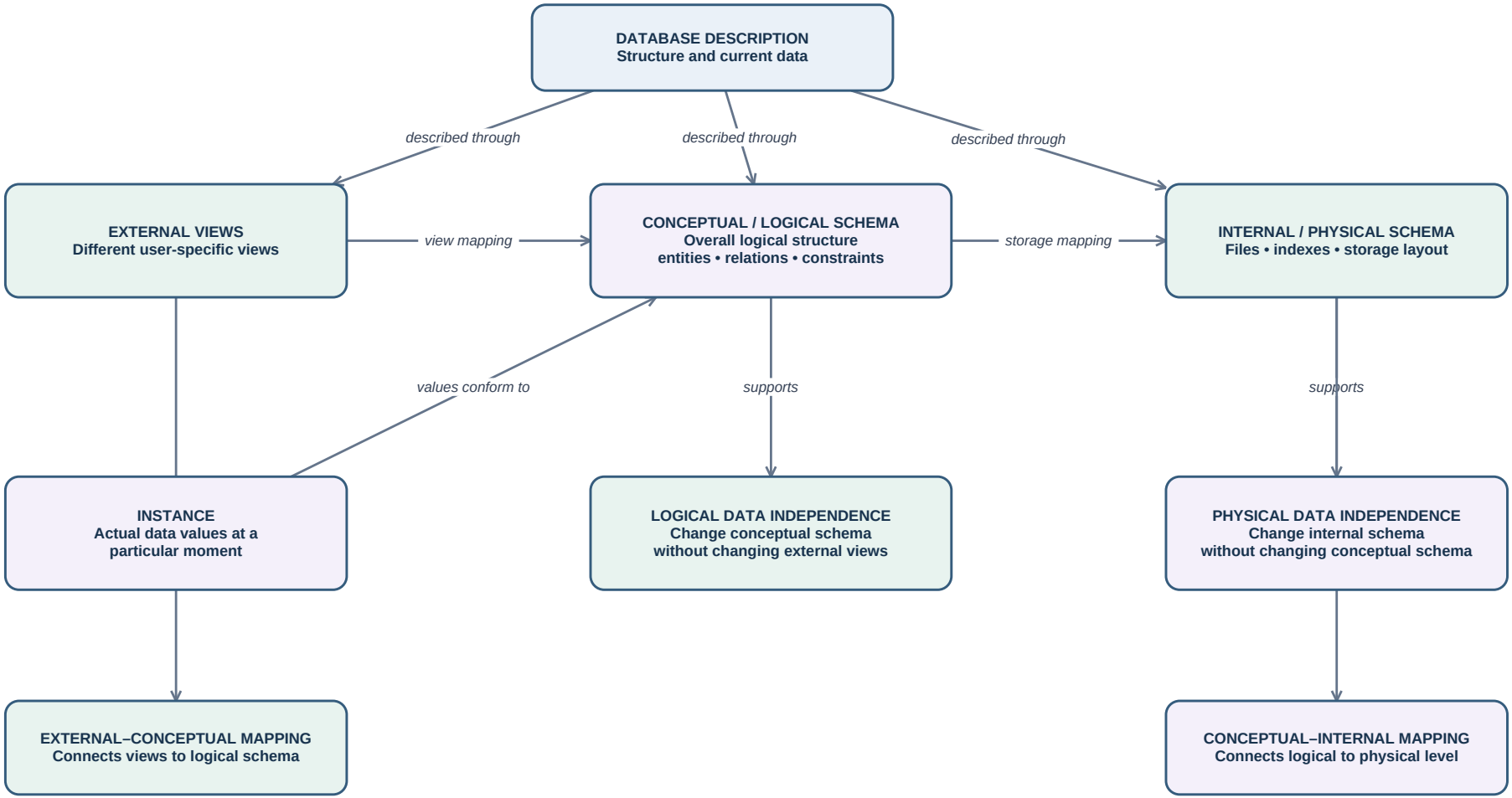



**How to explain the map:** Suitability comparison: Hierarchical is effective when data naturally forms a strict tree, but it is less flexible for many-to-many relationships. Network supports more complex linked records and many-to-many navigation, but its pointer-based structure can be complex. Relational organizes data into tables and uses primary/foreign keys and SQL, making it suitable for banking records and transactions. ER/EER is mainly a high-level conceptual design model: it helps identify entities, attributes, relationships and specializations before implementation as a database schema.

**Presentation tip:** Follow the arrow direction, read each linking phrase, and explain why the example fits the concept. Use the same labels when writing or presenting the map.

# Q4. SCHEMA, INSTANCE & DATA INDEPENDENCE

Three-schema view: changes at one level should not unnecessarily affect other levels

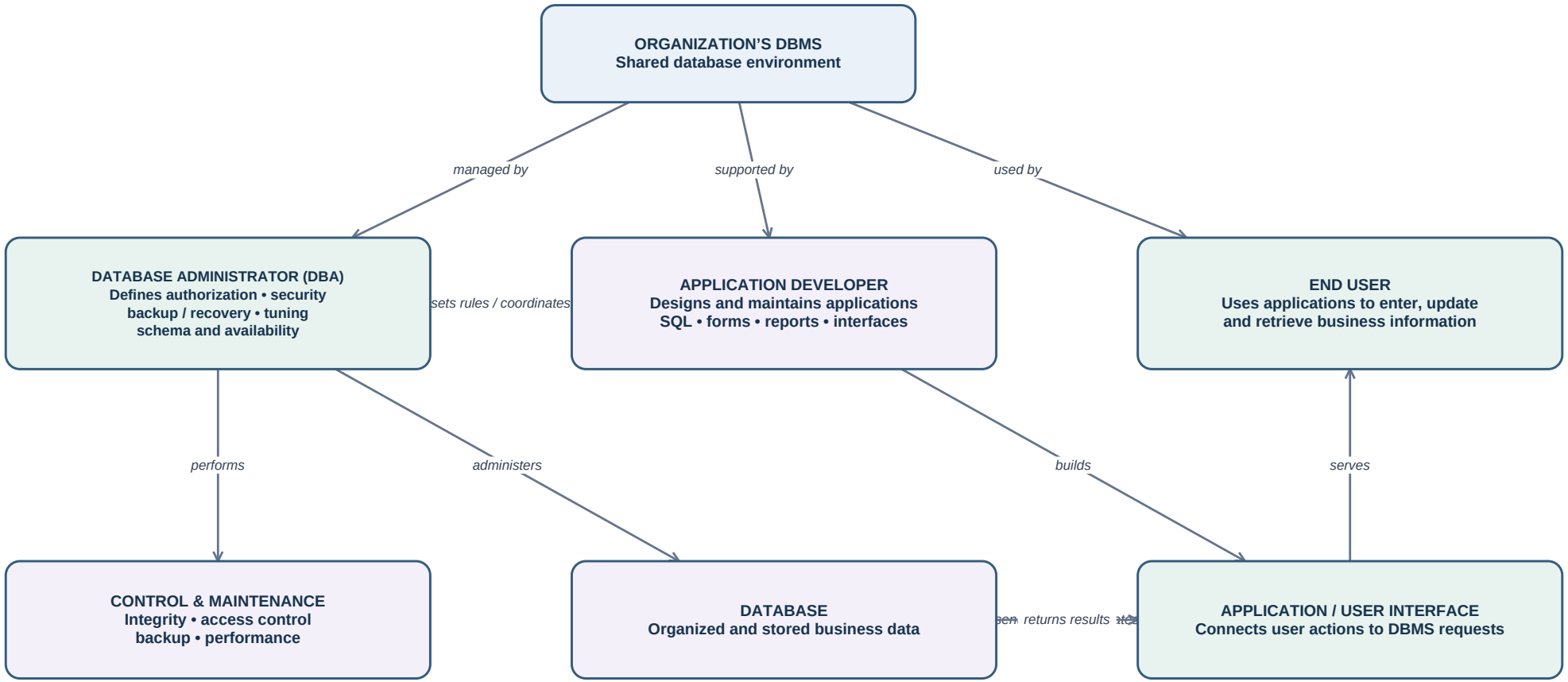



**How to explain the map:** Schema is the database's overall design/description; an instance is the actual data stored at a particular time. The conceptual (logical) schema describes what data is stored and the relationships/constraints. The internal (physical) schema describes how data is stored. Logical data independence means conceptual-schema changes can be made without requiring changes to external views or application programs. Physical data independence means storage-level changes can be made without changing the conceptual schema. The mappings connect the levels.

**Presentation tip:** Follow the arrow direction, read each linking phrase, and explain why the example fits the concept. Use the same labels when writing or presenting the map.

# Q5. ROLES IN THE DBMS ENVIRONMENT

How the DBA, end users and application developers contribute and interact




**How to explain the map:** The DBA is responsible for overall control of the database: user privileges, security, integrity, backup/recovery, performance and maintenance. Application developers build software that interacts with the DBMS using queries and interfaces. End users access data through applications to perform their tasks. Interaction: the developer creates the application, the end user uses it, the application sends requests to the DBMS, and the DBA manages and protects the database environment.

**Presentation tip:** Follow the arrow direction, read each linking phrase, and explain why the example fits the concept. Use the same labels when writing or presenting the map.